



Enhancing Medical Decision Support with Predictive Analytics and Uncertainty Quantification

Research Project

Research Question:

- Can CLUE provide feature-based uncertainty estimates to identify reasons for uncertain predictions?

Background:

- Underrepresented groups, such as people of colour, are not well represented in training datasets
- CDSSs should highlight uncertainties when such problems are encountered and notify the user (clinician)

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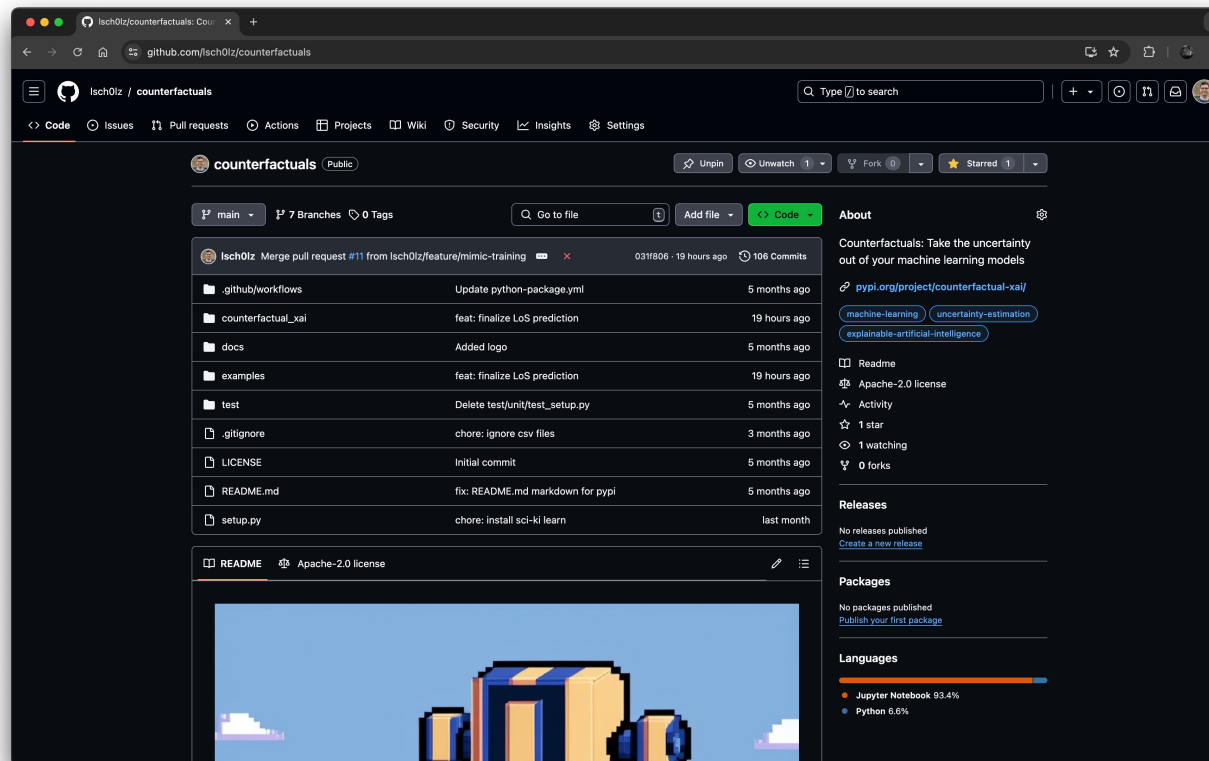
Goals:

- Re-implementation of the original CLUE repository in Python 3
- Train CLUE on medical use case (Length-of-Stay)
 - MIMIC III

Obstacles:

- Original CLUE repository was written in Python 2
- No clear documentation on how to use / train models

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Length-of-Stay Regression

- Predict how long a patient will stay in ICU
- LoS-Prediction is crucial for admission planning, resource allocation and care quality improvement [1]
- A lot of data cleaning required

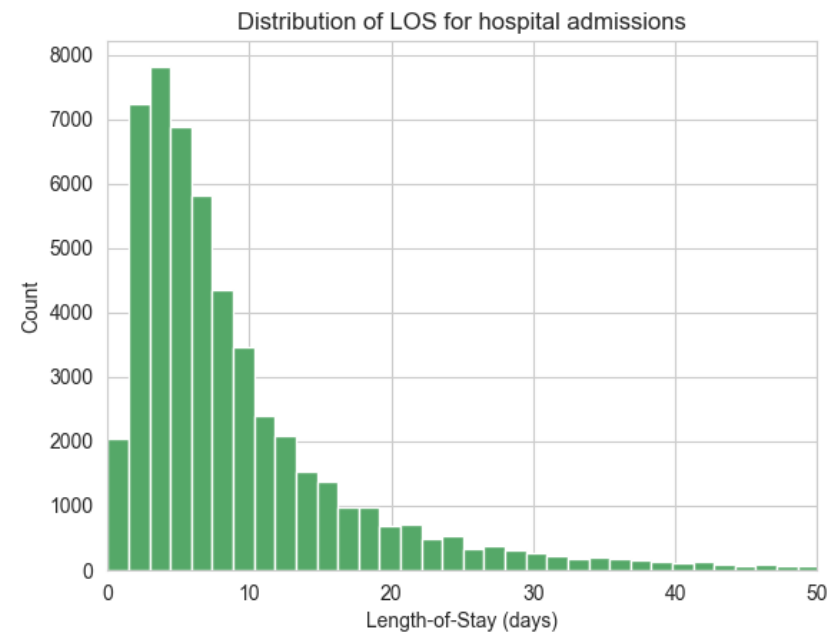
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LoS – Dataset [Credit: \[Link\]](#)

- MIMIC III
 - Admission, Diagnoses_ICD, ICU_Stays, Patients
- LoS = Admission / Discharge from hospital
 - Dead-People are dropped from dataset (LoS would be shorter)
- Categorize ethnicity, insurance- and marriage types
- Encode ICD-9 Codes into categories

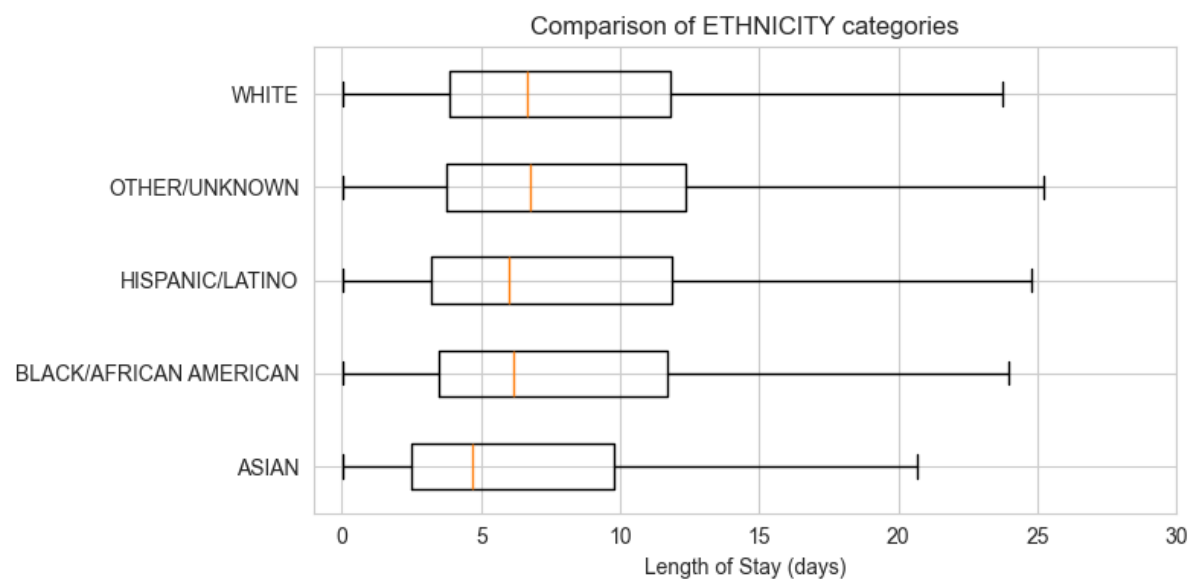
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LoS - Dataset



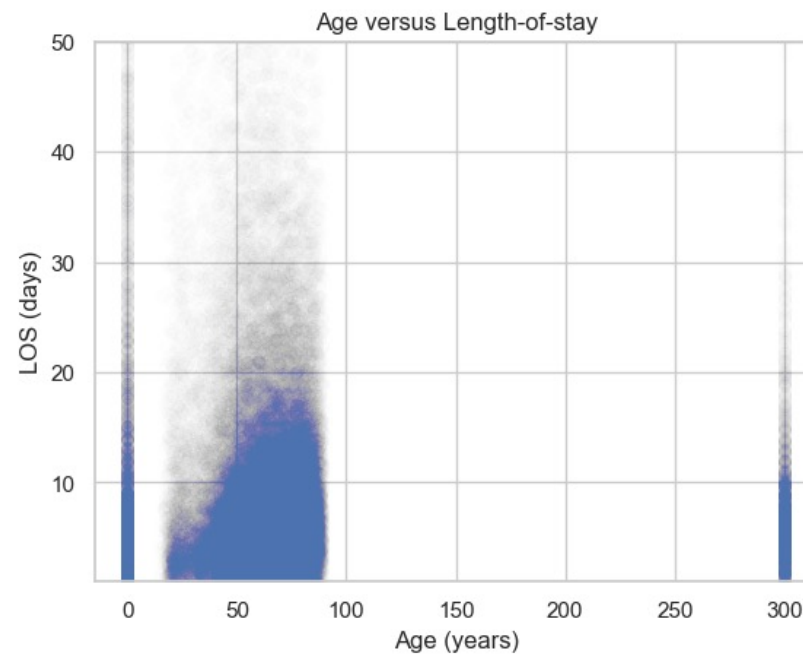
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LoS - Dataset



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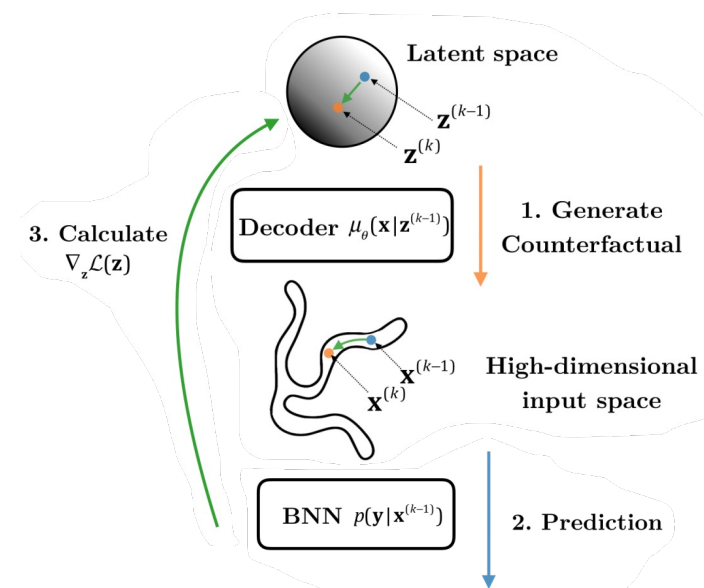
LoS - Dataset



CLUE [2]

- Train BNN, VAE on Dataset
- Input x_0 into trained VAE, to obtain z_0
- Produce Uncertainty Estimate through BNN $\mathcal{H}(y | \mu_\theta(x|z))$
- Backpropagate Gradient of $\mathcal{L}(z)$ and use Adam to optimise weights

Output: Uncertainty counterfactual $xCLUE$



Normal Uncertainty Quantization

Naive Approach:

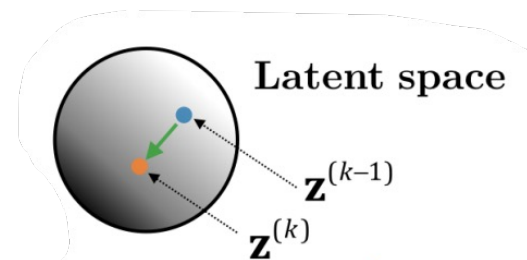
$$x_c = \underset{\text{counterfactual explanation}}{\operatorname{argmax}_x} (\overset{\text{predictor}}{p(y = c|x)} - \overset{\text{distance between } x \text{ and } x_0}{d(x, x_0)})$$

→ Adversarial outputs, that may be **not actionable**

→ Use DGM (VAE), to ensure explanations are in-distribution

CLUE - VAE

- VAE represents underlying data-structure
 - ensures generated counterfactuals are „in-distribution“
- Limits generation to more plausible data-points
 - Minimize adversarial outputs



CLUE - Objective

Uncertainty of BNN

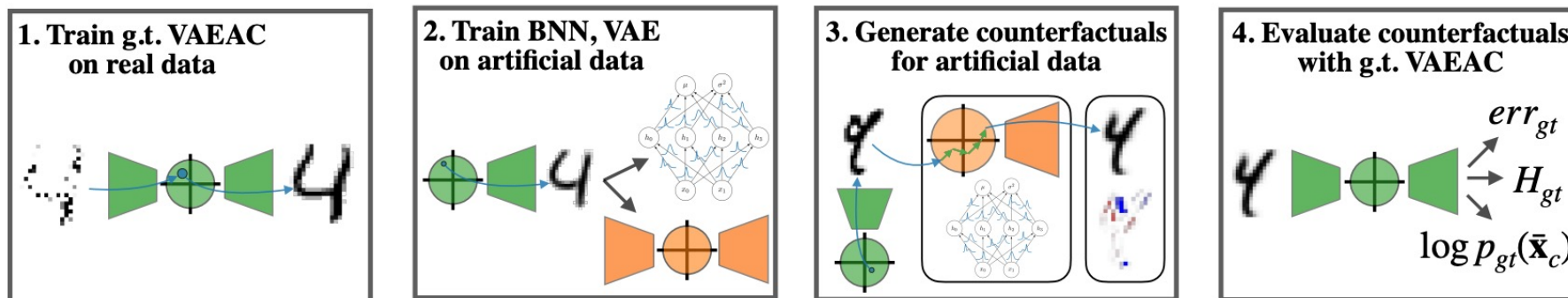
Distance of prediction and original input

$$\mathcal{L}(z) = \mathcal{H}(y \mid \mu_{\theta}(x|z)) + d(\mu_{\theta}(x|z), x_0)$$

- $x_{CLUE} = \mu_{\theta}(x|z_{CLUE})$
 - μ is prediction of BNN, z_{CLUE} is the latent variable
- $z_{CLUE} = \operatorname{argmin}_z \mathcal{L}(z)$
- Minimize uncertainty prediction of BNN, while keeping a close point to the original input x_0

CLUE – Evaluation Framework

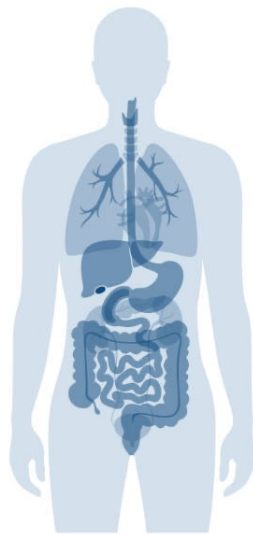
- BNN and VAE are trained on synthetic data
- Generate counterfactuals for uncertain samples
- Is generated counterfactual on-manifold of GT VAEAC?
- Does the model makes a lower prediction error through CLUE?



CLUE – LoS Regression

ICD-9 Categories

ICD-9 Counts



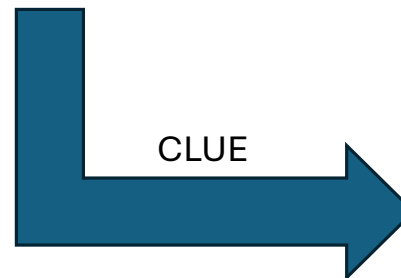
Name: Henry

Blood: 3
Circulatory: 4
Congenital: 0
Digestive: 5
Endocrine: 5
Genitourinary: 1
Infectious: 1
Injury: 8
Mental: 2
Misc: 5
Muscular: 0
Neoplasms: 0
Nervous: 0

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x_0 Into BNN:

LoS: 7.257 days



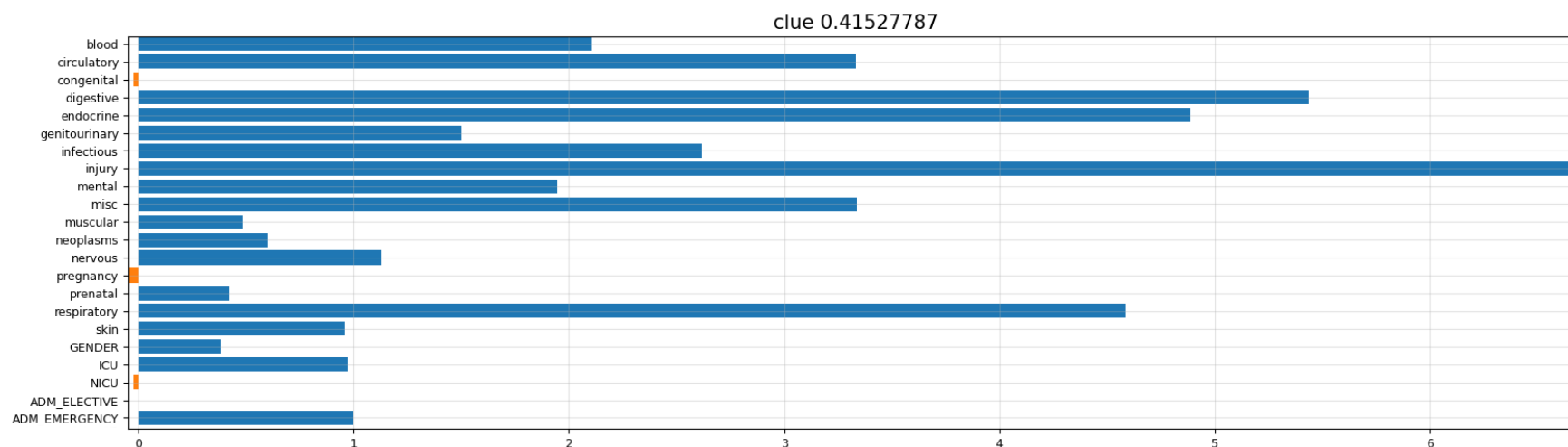
x_{CLUE} Into BNN:

LoS: 7.719 days

GT: 9 days

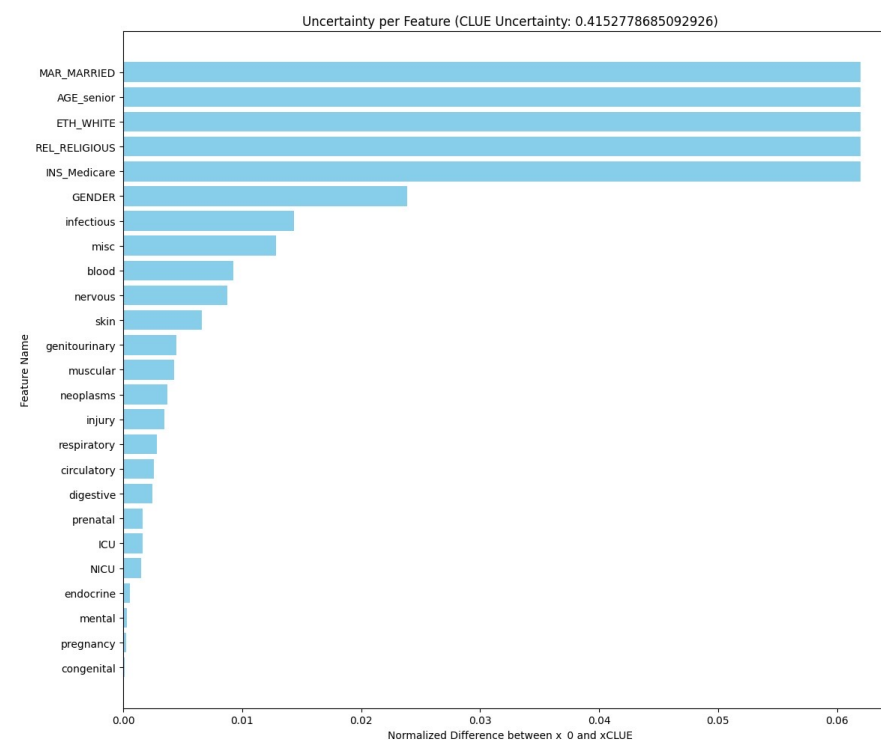
CLUE – LoS Regression

- Most uncertain prediction
- Improved values differ a lot from original values for amount of ICD-9 diagnosis



CLUE – LoS Regression [Contribution]

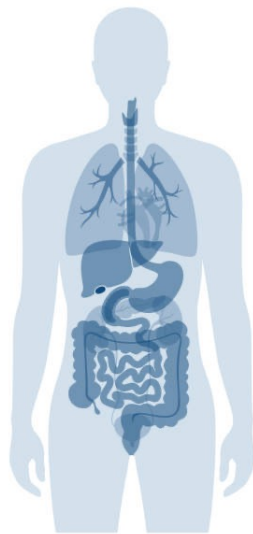
- Uncertainty per Feature is not available in original CLUE
- Normalize difference between x_{CLUE} and X_0 to obtain uncertainty per feature
- Difference = 1 → Value of categorical feature changed
- Example „AGE_senior“: Original patient is a senior, model would be more certain if he would **NOT** be a senior



CLUE – LoS Regression

ICD-9 Categories

ICD-9 Counts



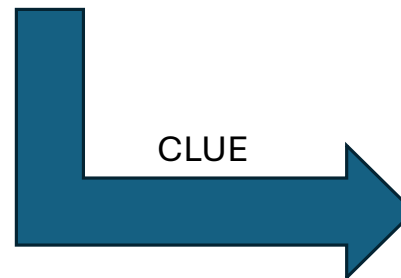
Name: Henrietta

Blood: 0
Circulatory: 2
Congenital: 0
Digestive: 1
Endocrine: 2
Genitourinary: 2
Infectious: 2
Injury: 8
Mental: 0
Misc: 2
Muscular: 2
Neoplasms: 0
Nervous: 5

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x_0 Into BNN:

LoS: 8.393 days



CLUE

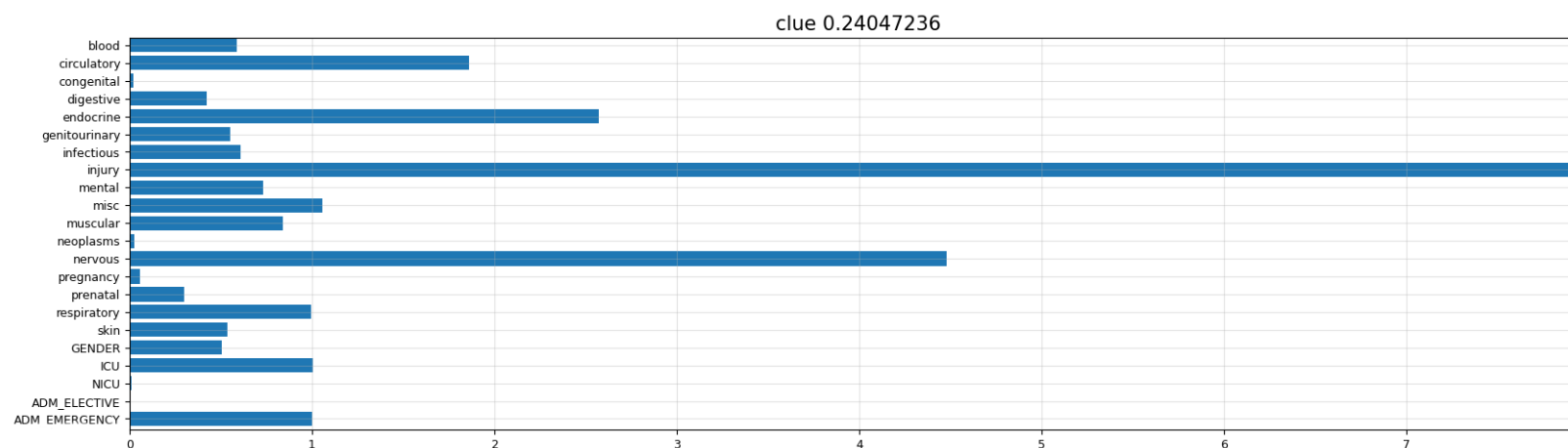
x_{CLUE} Into BNN:

LoS: 8.321 days

GT: 7 days

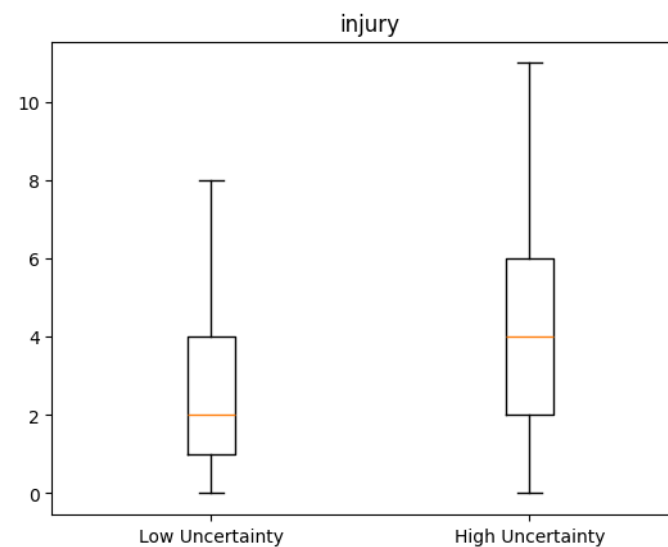
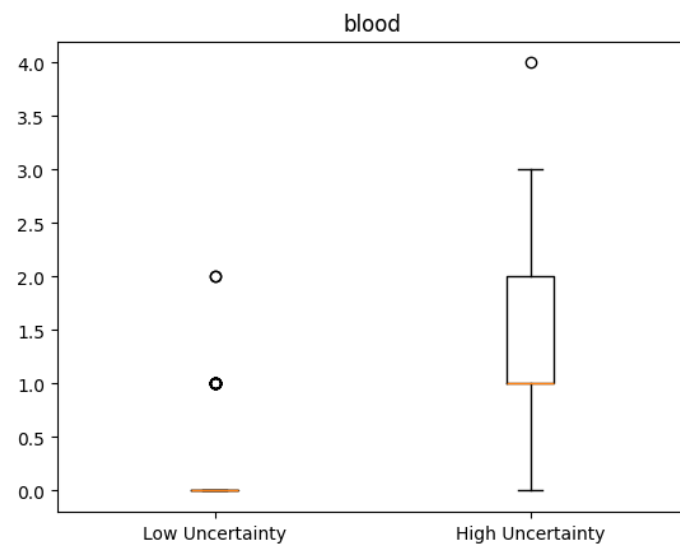
CLUE – LoS Regression

- Most certain prediction
- Improved values differ **not much** from original values for amount of ICD-9 diagnosis



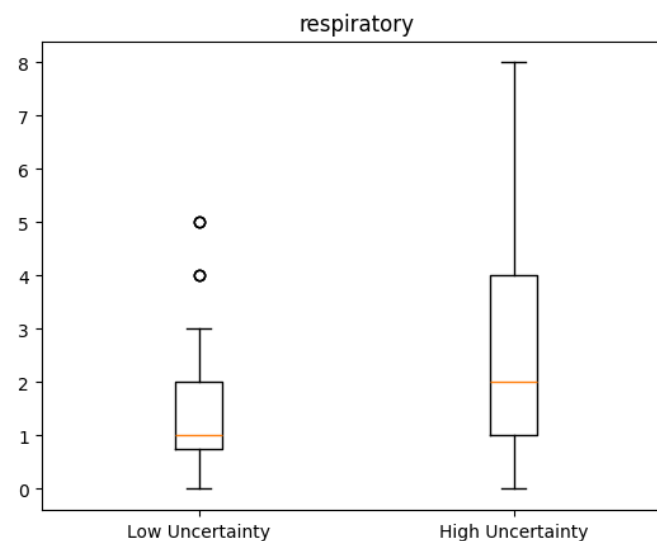
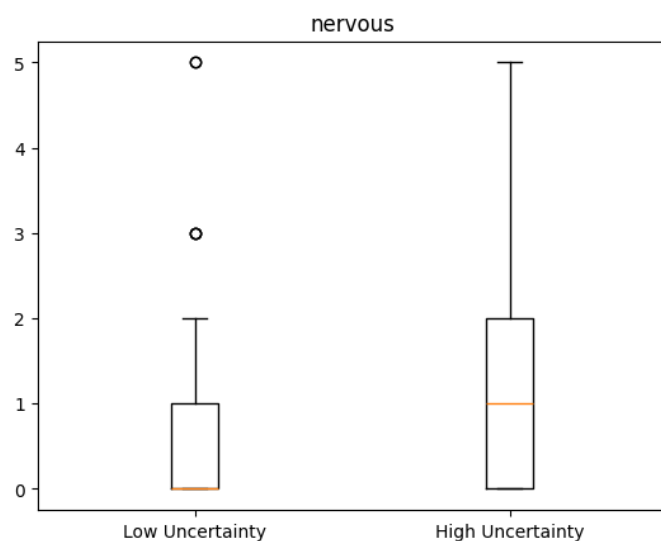
CLUE – LoS Regression

- Difference between Top-100 most certain and uncertain predictions in feature values
→ A lot of diagnoses lead to higher uncertainty



CLUE – LoS Regression

- Difference between Top-100 most certain and uncertain predictions in feature values



CLUE – LoS Regression

- Model is overall very certain about predictions
 - Blood, Gender, Nervous and Respiratory are the most different between certain and uncertain predictions
 - Uncertain predictions are all caused by more diagnoses in these ICD-9 categories
- More data points needed for cases with a lot of diagnoses